



MT1008 – Multivariable Calculus

Assignment No: 01

Individual Assignment

Section: All BSCS Sections

Semester: Spring 24

Date: 19-02-2024 before 10:00am

Marks: 190

Instructions:

1. Plagiarized work will result in zero marks.
2. No retake or late submission will be accepted.
3. Attach complete code, results, and screenshot for questions that require programming solution. Programs/codes should be typed.
4. **The assignment is to be submitted in softcopy as well as in hardcopy. Submit the hardcopy before the deadline through CR, and softcopy on GCR.**
5. The softcopy should be a single PDF file of your complete assignment including programming and non-programming questions.
6. The PDF file should be according to the following format: id_section_A1 e.g., i23-123456_A_A1. A1 in the end denotes Assignment 1. **The first page must include complete student information, including name, section, id, course name, and assignment number.**
7. The images of by-hand solution should be properly scanned. You can use any mobile application such as CamScanner or Adobe Scan for scanning. Each of these applications allow you to export pdf or image files which you can use to combine with your programming solutions. Do not attach direct images from the camera application of your mobile phone, or screenshots.
8. Python is the only approved programming language.

Question 1

A petrochemical company is designing a cylindrical tank with hemispherical ends to be used in the transportation of its products. Express the volume of the tank as a function of the radius r and length h of the cylindrical portion.

Question 2

An open box is to contain a volume of 15 cubic meters. Given that the material for the sides of the box costs \$3 per square meter and the material for the bottom costs \$6 per square meter, express the total cost C of the box as a function of length l and width w .

**Question 3**

A manufacturer estimates a production function to be $f(x, y) = 100x^{0.6}y^{0.4}$, where x is the number of units of labor and y is the number of units of capital. Use this function to show that if the number of units of labor and the number of units of capital are doubled, then the production level is also doubled.

Question 4

Find the domain of each function given below. Also graph the domain of functions with * symbol.

- i. * $f(x, y) = \frac{e^{\frac{1}{x}}}{\sin y}$
- ii. * $f(x, y) = \sqrt{\frac{x^2 + y^2}{x^2 - y^2}}$
- iii. * $f(x, y) = \sin^{-1} \frac{x^2}{y^2}$
- iv. * $f(x, y) = \frac{\ln x}{\ln y}$
- v. $f(x, y) = \arccos\left(\frac{y}{x}\right)$
- vi. * $f(x, y) = \arccos(y - x^2)$
- vii. $f(x, y, z) = \frac{z \sin x}{\cos y}$
- viii. $f(x, y, z) = \frac{\sqrt{4 - x^2} + \sqrt{1 - y^2}}{1 - \sqrt{9 - z^2}}$
- ix. $f(x, y, z) = e^{\sqrt{9 - (x^2 + y^2 + z^2)}}$

Question 5

The strength E of an electric field at a point (x, y, z) resulting from an infinitely long charged wire lying along the x – axis is

$$E(x, y, z) = \frac{3}{\sqrt{y^2 + z^2}}$$

Draw the surface using python. Also describe the level surfaces of E .

Question 6

The formula $V(x, y) = \frac{9}{\sqrt{4 - (x^2 + y^2)}}$ gives the electric potential V at a point (x, y) in xy – plane. Draw the equipotential curves for $V = 18, 9$, and 6 volts. Describe the surface $Z = V(x, y)$.

Question 7

Sketch the level curves of the function for given C -values; $C = 0, 1, 2, 3$.

$$f(x, y) = \begin{cases} \sqrt{x^2 + y^2} & x \geq 0 \\ |y| & x < 0 \end{cases}$$



Question 8

Specify the interior and the boundary of the set. State whether the set is open, closed or neither. Then sketch the set.

- a) $[(x, y): 4 < x^2 \leq 9]$
- b) $[(x, y): y \leq x^2]$

Question 9

Check if the limit exists:

- a) $\lim_{(x,y) \rightarrow (2,e)} (x^2 y \ln y)$
- b) $\lim_{(x,y) \rightarrow (3,2)} \frac{y^2 - 4}{xy - 2x}$
- c) $\lim_{(x,y) \rightarrow (0,0)} \left(\frac{x^2 - y^2}{x^2 + y^2} \right)^2$
- d) $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 y^2}{x^4 + y^4}$

Question 10

- a) Determine whether each function f is continuous at $(0,0)$.
- b) If f is discontinuous at $(0,0)$, is it possible to define $f(0,0)$ so that f would be continuous at $(0,0)$?
- c) If the answer to part b is yes, how should $f(0,0)$ be defined?
- i. $f(x, y) = \frac{xy^2}{x^2 + y^2}$
- ii. $f(x, y) = \begin{cases} \frac{3xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$

Question 11

Find each limit by converting to polar coordinates.

- a) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$
- b) $\lim_{(x,y) \rightarrow (0,0)} \frac{\cos(x^2 + y^2)}{x^2 + y^2}$

Question 12

Determine where each function is continuous.

- a) $f(x, y) = \sin(x + y) \cos(x - y)$
- b) $f(x, y, z) = \frac{\sin z}{e^x + e^y}$
- c) $f(x, y) = \frac{x + 3xy^2}{e^{x^2 - y^2}}$
- d) $f(x, y) = \frac{\ln(x - y)}{(x^2 + y^2)^{\frac{3}{2}}}$

**Question 14**

The temperature distribution of a heated plate at a point (x, y) in the xy – *plane* is modeled by $T = T(x, y) = \frac{100}{\sqrt{x^2 + y^2}}$, $1 \leq x^2 + y^2 \leq 9$.

Find the rate of change of T in the direction of the positive x – *axis* at the point $(1, 0)$ and at the point $(0, 1)$.

Question 15

The temperature distribution T of a heated plate at a point (x, y) in the xy – *plane* is modeled by

$$T = T(x, y) = \frac{100}{\ln 2} \cdot \ln(x^2 + y^2) \quad 1 \leq x^2 + y^2 \leq 9$$

Find the rate of change of T in the direction of positive y – *axis* at the point $(2, 0)$ and point $(0, 2)$.

Question 16

Show that the function

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

has partial derivatives at $(0, 0)$, but is continuous at $(0, 0)$.

Question 17

A manufacturer wants to make an open rectangular box of volume $V = 500 \text{ cm}^3$ using the least possible amount of material. Find the dimensions of the box.

Question 18

The temperature T in degrees Celsius at any point (x, y) of a flat plate in the xy – *plane* is $T = 60 - 2x^2 - 3y^2$. Use python to draw the isothermal curves (level curves) for $T = 60^\circ\text{C}$, 54°C , 48°C , 6°C , and 0°C . Describe the surface $z = T(x, y)$.

Question 19

Compute the partial derivatives of the following functions using python:

- a) $f(x, y) = x^2 + 2xy + y^3$
- b) $f(x, y, z) = x^3 + 2xyz - y^2z$
- c) $h(x, y) = e^{xy} \cos(x + y)$